

## 기계열역학 (공) (MECH204-01)

|      |                                                |
|------|------------------------------------------------|
| 학기   | 2026년 2학기                                      |
| 담당교수 | 김민수                                            |
| 이수구분 | 교선평수                                           |
| 학점   | 3.0                                            |
| 강의시간 | 화10:30-12:00(E7 - G02), 목10:30-12:00(E7 - G02) |
| 강의실  | E7 207 (교수실)                                   |
| 소속   | 기초학부                                           |

### 교과개요

Mechanical engineers require a solid foundation in thermodynamics and the ability to analyze thermal systems for engineering design and problem-solving. This course introduces the fundamental principles of engineering thermodynamics and their applications to energy conversion systems. Topics include thermodynamic properties, energy and mass balances, the first and second laws of thermodynamics, entropy, control volume analysis, gas power cycles, vapor power cycles, refrigeration systems, and heat pump systems. A simple thermal engineering project will be conducted to reinforce theoretical concepts through practical engineering applications

### 교과목표

Upon successful completion of this course, students will be able to: 1) Understand and apply the fundamental principles of engineering thermodynamics. 2) Analyze closed and open systems using the first and second laws of thermodynamics. 3) Evaluate thermodynamic properties and entropy changes in engineering processes. 4) Analyze and compare gas power cycles, vapor power cycles, refrigeration systems, and heat pump systems. 5) Develop physical intuition and problem-solving skills through practical engineering examples and applications.